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# Bacteria

Bacteria can be found in almost any environment and around 2000 species have been identified. They are tiny and their sizes are usually described in μm (micrometres = $10^{-6}$ m).

**1** What is the singular form of *bacteria*?

There are several ways to classify bacteria, one of the simplest is to categorise them by their shape:

- cocci (s. *coccus*) – spherical
- bacilli (s. *bacillus*) – rod shaped
- spirilla (s. *spirillum*) – spiral or curved.

Cocci are spherical bacteria. They can exist as single or clumped cells and cannot move themselves – they are not motile. Diplococci are paired cells (*di* two) and staphylococci form clusters (from Gk. *staphyle* bunch of grapes, which is what they resemble). Streptococci are long chains (Gk. *streptos* curved).

Identify the types of bacteria shown below, based on the descriptions given.



**2** _____ **3** _____ **4** _____ **5** _____

**6 Research:** Find a disease caused by each type of coccal bacteria.

Bacilli may possess flagella (s. *flagellum*), small hair-like appendages that move to provide locomotion for the bacteria (the name flagellum is derived from the Latin word meaning 'whip'). Spirilla (from the Latin meaning 'coil') are long curved bacteria. They sometimes have tufts of flagella.




**7** What use does a bacillus make of a flagellum?

**8** How many micrometres are there in a metre?

**9** Cocci are described as 'not being motile'. What does this mean?

**10** Identify one way of classifying bacteria.

## Structure of bacteria

Bacteria are procaryotes (variation prokaryotes); they lack a distinct nucleus. The cell is made up of a cell wall and cell membrane that encloses cytoplasm. The DNA, containing genetic information, is a single molecule in direct contact with the cytoplasm rather than being enclosed in a nucleus. Bacteria are thought to have evolved around 2 billion years ago.

**11** Why might it be difficult to decide when bacteria evolved?

**12** What is the shape of spirilla?

**13 Research:** Give a definition for *spirochete*.

The cell wall gives the bacteria its shape and strength. The cell (plasma) membrane is semipermeable; it controls the movement of substances in and out of the cell.

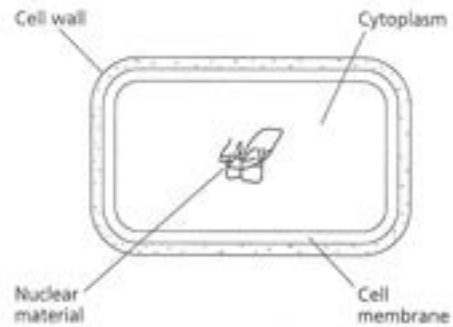


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**14** Define *semipermeable*.

The cytoplasm is a watery gel containing organic and inorganic substances.

Simplified diagram of bacterium



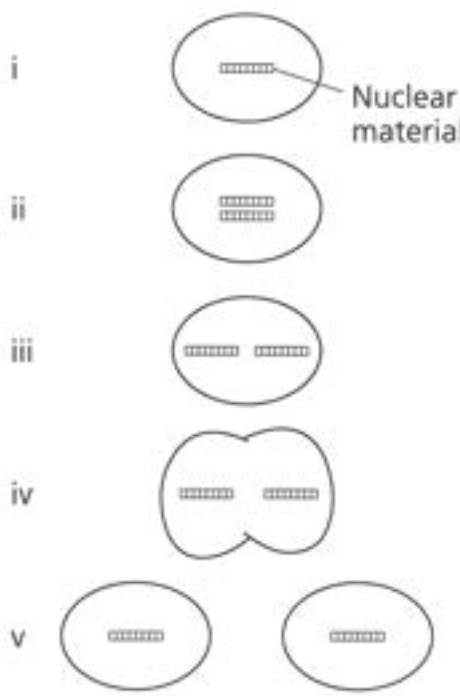

## Reproduction of bacteria

Bacteria reproduce asexually by binary fission. This is a process whereby an organism divides into two parts, each of which develops into two (usually genetically identical) organisms – sometimes called 'daughter' cells.

**15** Outline the differences between sexual and asexual reproduction.

### Binary fission

A simplified model of binary fission is shown in the diagram. (Binary *two*, fission *to split*)




**16** Rearrange the following statements so that they are in the right order.

**A** Two daughter cells are formed, each having a nuclear area.
**B** A bacteria cell has a single chromosome made up only of DNA.
**C** The chromosomes split apart.
**D** The cell membrane moves inward to partition the cytoplasm.
**E** The chromosome is replicated.

Correct order is: _____

Compared to more complex organisms, bacteria have the potential to reproduce extremely quickly. It is not uncommon to find bacteria that, under favourable conditions, can undergo cell division (replicate) at intervals of 20 to 40 minutes.

**17** A species of bacteria replicates every 30 minutes. Starting with a single cell, calculate how many bacteria could exist after 24 hours if this rate of replication was maintained (show working). Explain why, in practice, this number would be unlikely to be achieved.

**18** Draw up a table that shows the major parts, and their functions, of a bacterium cell.

**19** Read the following passage and give definitions for the terms in italics. You may need the help of a dictionary or text book.

Bacteria are found in practically all types of *environments*. Some types of bacteria exist in icy conditions and others near *hydrothermal vents* deep in the ocean. Some bacteria can exist in the presence of oxygen and others are *anaerobic*. Bacteria *metabolise* a wide range of materials. Gram's stain is one test used to *differentiate* between various types of bacteria. One way to classify bacteria is to group them as *symbionts* or *saprophytes*. Bacteria can affect foodstuffs in many ways – either to make them *unpalatable* or to preserve them. Some bacteria are *pathogenic* and scientists have developed *antibiotics* to use against these bacteria.

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